

Fig. 2: Plasma-generation technique

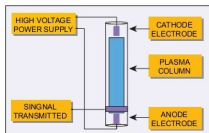


Fig. 3: Generating a low-temperature plasma with electric field application

behaves as a fluid without net charge.

(ii) *Collectively*. Meaning that plasma as a whole is capable of carrying out processes that generate electric and magnetic fields to which it can react. This is one of the most important properties that lead to some unparallel characteristics.

So basically, plasma is produced "when enough atoms are ionised to significantly affect the electrical properties of a gas under normal conditions." It is not any alien form but only a state of ionisation, a state of matter. There are eleven elements that exist as gas under normal conditions. Out of these, Group 18 elements are of prime importance (if we are analysing discharge tubes) as they are inert. Plasma is much visible matter in the universe, being about 99 per cent of all the matter. Besides astronomical plasma, we can distinguish two main groups of plasma from laboratory point of view:

(i) *Thermal plasma*. Here electrons and ions are at thermal equilibrium. But for this thermal equilibrium to exist, very high temperatures are required, specifically in the range of 4000K-20,000K. But this equilibrium

makes this kind of plasma to be unfit for the antenna system application (welding, plasma torches, sintering and etching).

(ii) *Non-thermal plasma*. Here ions and neutral particles are at lower temperature as compared to electrons, or we can say that the electrons are somewhat 'hotter' than ions. This fits for the antenna system application.

Fluorescent lamps and neon signs are examples of plasma states. In case of lamps, the extra electrons added at the electrode ionises the gas with the help of mercury atoms and the result is ionised state of plasma.

The most common way of generating a low-temperature plasma is by applying electric field to a neutral gas. The neutral gas always contains a few electrons and ions. When an electric field is applied to it, these free carriers are accelerated and they collide with atoms and molecules. This is similar to an avalanche breakdown, that is, a large number of charge carriers are formed at the loss of old carriers, and so there is a balance in carrier generation. In this way, steady-state plasma is developed.

Consider DC discharge system for a rough idea. If we subject the gas to high pressure, there would be a high-collision rate and this could produce thermal plasma. But DC discharge system requires non-thermal plasma, so we will pressurise the gas at low pressure to produce a low-collision rate. A high electric field is applied to the electrode plates and the phenomena stated above takes place. Following

this pattern, if we apply more voltage, we will get more plasma density. But if we apply electric current in microsecond pulses, we can obtain high plasma density at the same input power. This can be advantageous as the voltage requirements of the system are determined by the nature of gas, electrode material, current density and length of the tube.

Earlier, the DC discharge systems were very bulky but now engineers have been able to reduce their size. But the ionisation instrument is still bulky and hence it limits its universal application.

Interaction of plasma with EM waves

As we are interested in designing antenna systems, the interaction and behaviour of plasma with EM waves must be investigated for better understanding. As per some theses, various relations are set up (qualitatively) to study their interaction. Plasma contains quasi-neutral particles, which means they are highly conductive. For analysing interaction with EM waves, four important parameters are conductivity, electrical permittivity, magnetic permeability and propagation constant.

Qualitatively, the interaction of plasma with EM waves can be formulated as:

1. Plasma with high-collision frequency behaves as a lossy medium.

This is due to the reason that with increase in plasma pressure and electron density, the rate of collision increases, so considerable amount of energy will be lost. (This is the reason why thermal plasma is unfit to be used in antennae.)

2. If $W > W_p$, EM wave frequency is greater than plasma frequency (an inherent property of plasma), so wave propagates in plasma and the plasma has dielectric properties which are electrically controllable.

3. If $W < W_p$, The propagation constant is imaginary. The wave is vanishing with the plasma medium. The wave can be absorbed or reflected depending on the collision frequency.

4. *Effect of magnetic field*. The